MILITARY SPECIFICATION
RECEIVER-TRANSMITTER, LASER: 10559281

WARNING - EYE HAZARD

EXTREME CAUTION MUST BE OBSERVED IN OPERATING AND TESTING THE RECEIVER-TRANSMITTER. IF THE LASER LIGHT BEAM ENTERS THE HUMAN EYE, THE RETINA OF THE EYE CAN BE BURNED. THE HIGH ENERGY DENSITY OF THE BEAM CAN ALSO PRODUCE SUCH DAMAGE AT EXTREME RANGE AND BY REFLECTION FROM ANY SHINY SURFACE SUCH AS GLASS, METAL, PAINT, ETC. IN VIEW OF THE POSSIBILITY OF PRODUCING SERIOUS EYE DAMAGE, THE RECEIVER-TRANSMITTER SHOULD NOT BE OPERATED UNLESS THE TARGET AREA HAS BEEN EVACUATED.

1. SCOPE

1.1 This specification covers one type of electro-optical unit known as Receiver-Transmitter, Laser: 10559281 (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-C-14880	Component Assembly: 10559479 (Transmitter)
MIL-F-13926	Fire Control Materiel: General Specification Governing the Manufacture and Inspection of
MIL-P-14232	Parts, Equipment and Tools for Army Materiel: Packaging and Packing of
MIL-I-45607	Inspection Equipment, Supply and Maintenance of

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STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109	Quality Assurance Terms and Definitions
MIL-STD-810	Environmental Test Methods

DRAWINGS

U.S. Army, Frankford Arsenal

F10559281	Receiver-Transmitter, Laser

PACKAGING DATA SHEET

10559281	Packaging of Receiver-Transmitter, Laser: 10559281

(Copies of specifications, standards, drawings, and packaging data sheets required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Fabrication. - The receiver-transmitter shall be manufactured in accordance with Drawing F10559281 and drawings pertaining thereto.

3.2 General specification. - The receiver-transmitter shall meet the following requirements, where applicable, of MIL-F-13926:

(a) Order of precedence
(b) Dimensions and tolerance
(c) Part identification and marking

3.2.1 Electronic workmanship. - The receiver-transmitter shall meet the workmanship requirements of Drawing F10559281 and drawings pertaining thereto.

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3.3 Performance.- Unless otherwise specified, the receiver-transmitter shall meet the performance requirements of this specification at standard ambient temperatures between 60 degrees Fahrenheit (^{o}F) and 90°F.

3.3.1 Loads, power and signals.- The receiver-transmitter shall perform as specified herein when the loads and auxiliary equipment of Table I are connected as specified, and the inputs of Table I are applied.

TABLE I

ITEM	CONDITION	CHARACTERISTIC	CONNECTIONS
1.	Loads:		Unless otherwise specified, connected to the following pins with return to J1-T:
1.1	Resistor	300 ohms ±5%	J1-A
1.2	Resistor	300 ohms ±5%	J1-B
1.3	Resistor	300 ohms ±5%	J1-C
1.4	Resistor	300 ohms ±5%	J1-D
1.5	Resistor	300 ohms ±5%	J1-E
1.6	Resistor	300 ohms ±5%	J1-F
1.7	Resistor	300 ohms ±5%	J1-G
1.8	Resistor	300 ohms ±5%	J1-H
1.9	Resistor	300 ohms ±5%	J1-J
1.10	Resistor	300 ohms ±5%	J1-K
1.11	Resistor	300 ohms ±5%	J1-L
1.12	Resistor	240 ohms ±5%	J1-P
1.13	Resistor	620 ohms ±5%	J1-R
1.14	Resistor	20 kilohms ±5%	J4-r
1.15	Resistor	750 ohms ±5%	J4-q (return to J4-W)
1.16	Resistor	1 kilohm ±5%	J4-p (return to J4-W)
1.17	Resistor	27 kilohms ±5%	J4-U (return to J4-X)

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TABLE I (continued)

ITEM	CONDITION	CHARACTERISTICS	CONNECTIONS
2.	Power sources:		Connected between:
2.1	+28 V Tank Power	24 ±2 Vdc	J2-J(+) and J2-L(-)
2.2	+24 V Auxiliary	24 ±2 Vdc	J2-E(+) and J2-L(-)
2.3	24 V Q-switch	24 ±2 Vdc	J2-A(+) and J2-L(-)
3.	Signal sources:		
3.1	Laser On	24 ±2 Vdc	Applied between J1-M(+) and J1-T(-) unless otherwise specified.
3.2	Remote Range	8 ±0.5 Vdc in series with 33 kilohms ±5%	Applied between J1-U (+) and J1-T(-) when specified herein.
3.3	Gunners Control	24 ±2 Vdc	Applied between J1-N(+) and J1-T(-)
3.4	Digital Type A	Logical one: 4 ±1 V Logical zero: 0.2 ±0.2V	Applied to input terminals as specified herein.
3.5	Digital Type B	Logical one: 15 ±1 V Logical zero: Open circuit	Applied to input terminals as specified herein.
3.6	Turn-on Reset	See Figure 1	Applied as specified on Figure 1 whenever the Laser On signal is applied.
4.	Auxiliary equipment:		
4.1	Circuit Card Assembly: 10559205 (Indicator, Command, Display) or equivalent		Connected as shown on Figure 2.
4.2	Range Data Output Generation Circuit	See Figure 3	Connected as shown on Figure 3 when specified herein.

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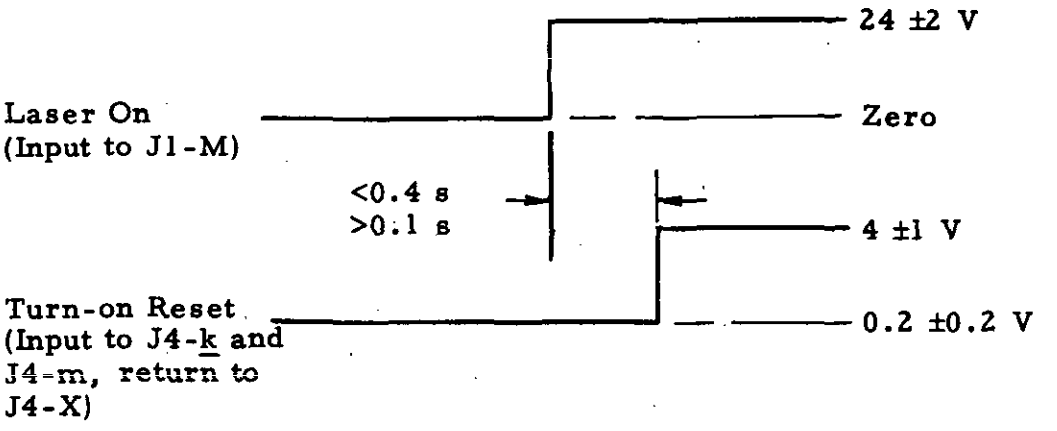


Figure 1. Turn-on Reset Time

3.3.1.1 Power controls.

3.3.1.1.1 Laser On.- With the digital type A logical zero signal of Table I applied to J4-s and J4-h, followed by the application of the Laser On signal as specified in Table I, the specified d-c voltages shall appear at the terminals listed in Table II. The waveform of Figure 4 shall appear at the indicated terminal within 6 seconds after application of the Laser On signal.

TABLE II

OUTPUT PINS	OUTPUT VOLTAGES
J2-F	15 ±2 Vdc
J2-G	29 ±3 Vdc
J2-H	24 ±2 Vdc
J4-U	235 ±35 Vdc
J4-V	15 ±2 Vdc
J4-N	5 ±1 Vdc

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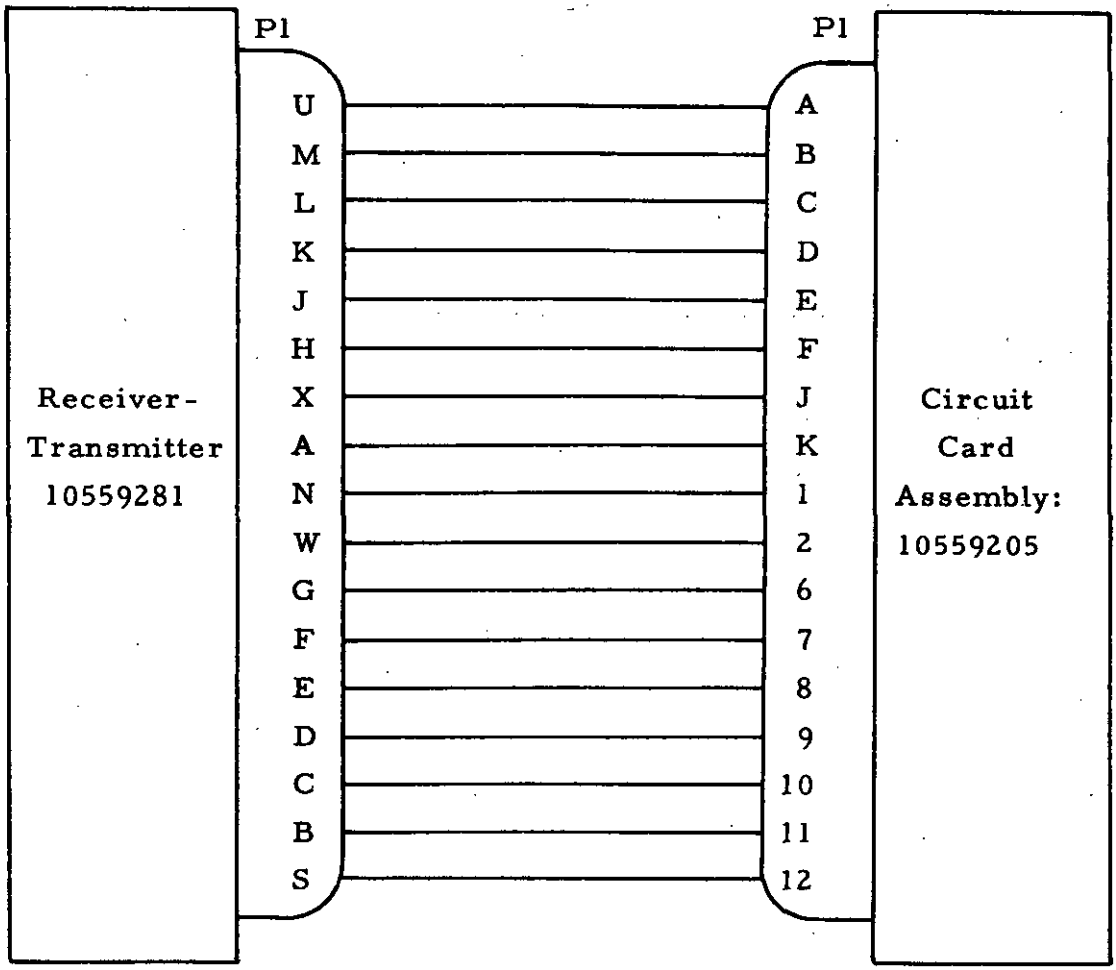
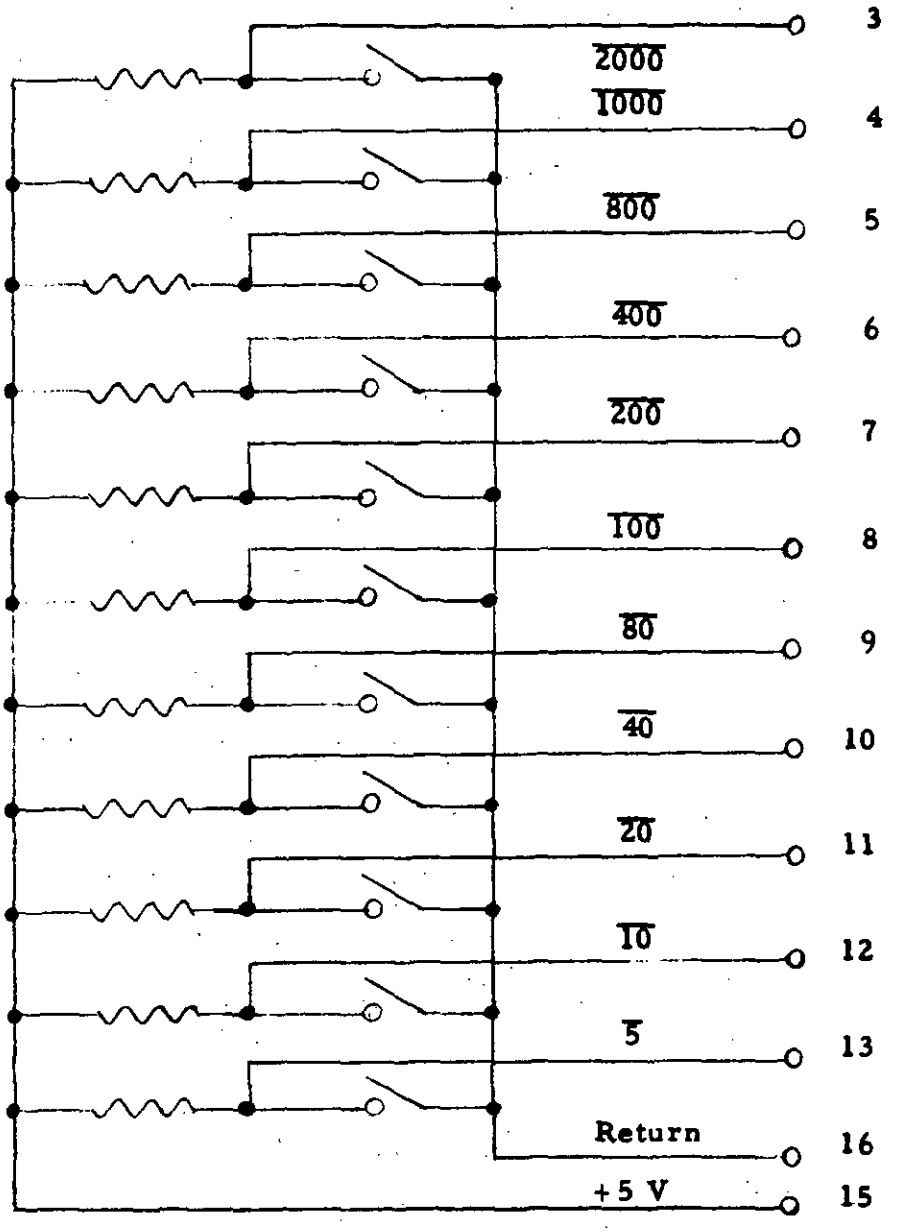


Figure 2. Range Indicator Interconnections

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NOTES: 1. All resistors are 2 kilohms ±5%, 1/4 W

2. Pin numbers on Receiver-Transmitter schematic, counters card connector A1XA5

Figure 3. Range Data Output Test Schematic

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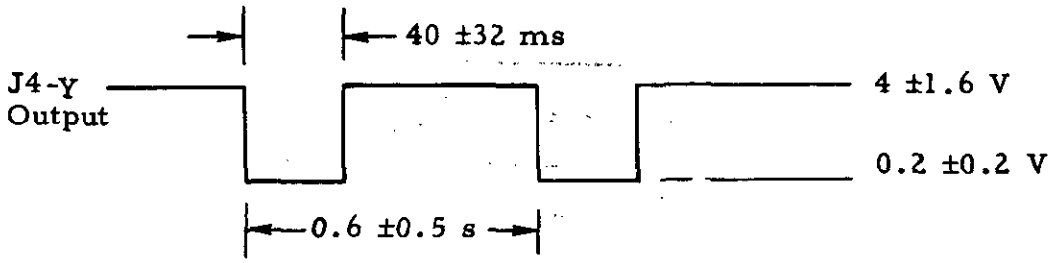


Figure 4

3.3.1.1.2 Safety interlock.- When the receiver-transmitter cover is removed, switch S-1 shall activate and turn the unit off. The voltage at J2-H shall be 0.0 ±0.5 Vdc.

3.3.1.1.3 Emergency power switch.- With the Laser ON and gunners control signals not present and the emergency power switch sequenced through the following positions; OFF, COMMANDERS CONTROL, GUNNERS CONTROL, the following shall occur:

(a) OFF	Power off, 0 ±0.5 Vdc at J2-H
(b) COMMANDERS CONTROL	K-1 shall be activated; 24 ±2 Vdc shall appear on J2-H; zero ±0.5 Vdc shall appear on J4-r.
(c) GUNNERS CONTROL	K-1 shall be activated; 24 ±2 Vdc shall appear on J2-H. The voltage at J4-r shall be 23.5 ±3 Vdc.

3.3.1.1.4 Low voltage.- When the 24 Vdc primary power is removed from J2-J, the voltage at J3-J shall remain at 15 ±2 Vdc.

3.3.1.1.5 Gunners control.- With 24 ±2 Vdc applied to J1-N, the voltage at J4-r shall be 23.5 ±3 Vdc.

3.3.1.2 Data line (to computer).- With the circuit indicated on Figure 3, or equivalent, substituted for the counters circuit card (10559305), the output at J1-A through J1-L shall be as follows:

(a) Switches open	Zero ±1.0 volt
(b) Switches closed	7.5 ±0.5 volts

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NOTE: The symbols used to represent digital signals in Tables III through IX are defined as follows:

1 denotes logical one, type A, per Table I
0 denotes logical zero, type A, per Table I
C denotes logical one, type B, per Table I
D denotes logical zero, type B, per Table I
H denotes 4.0 ±1.6 volts
L denotes 0.2 ±0.2 volt

3.3.1.3 Target selection and reset.- With the type A and type B signals of Table I applied to the group of input pins for each item as specified in Table III, and each item performed in sequence, a digital signal shall appear at each listed output pin and the range readout specified in Table III shall be obtained for that item.

3.3.1.4 Command signal output (to computer).- With the digital type A and type B signals of Table I applied to each group of input pins for each item as specified in Table IV, and each item performed in sequence, a digital signal shall appear at each output pin listed in Table IV for that item.

3.3.1.5 Test access output.- With the test oscillator enabled (large cover removed) and the digital type A and type B signals of Table I applied simultaneously to the input pins as specified in Table V, followed by a logical zero type A signal momentarily applied to J4-k and J4-m, the output waveforms or voltages specified in Table V shall appear at the indicated pins of J3.

3.3.1.5.1 AGC inhibit.- Under the above conditions and with the receiver-transmitter window covered, the d-c voltage at J3-D shall be 0 ±1 Vdc.

3.3.1.6 PFN charge and laser firing.- With the digital type A and type B signals of Table I applied to each group of input pins for each item as specified in Table VI, and each item performed in sequence, the output at J4-Y shall be as specified in Table VI for that item.

3.3.1.7 Laser mode control.- With the digital type A and type B signals of Table I applied to each group of input pins for each item as specified in Table VII, and each item performed in sequence, the digital outputs at J4 and the range (nixie) readout shall be as specified in Table VII for that item.

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TABLE III

ITEM	INPUT PINS OF J4									OUTPUT PINS OF J4						RANGE READOUT 3/ (NIXIES) (±10 meters)
	h	e	f	g	j	k	m	n	s	P	R	S	T	p	q	
1	0	0	0	1	C	1 1/	1 1/	0	0	H	L	L	H	H	H	0000
2	0	0	0	1	C	1	1	0	0 2/	H	L	L	L	H	L	0830
3	0	0	0	1	C	1	1	0	0 2/	L	H	L	L	H	L	1835
4	0	0	0	1	C	1	1	0	0 2/	H	H	L	L	H	L	2830
5	0	0	0	1	C	1	1	0	0 2/	L	L	H	L	L	H	2830
6	0	1	0	0	C	1	1	0	0	L	L	H	L	L	H	0830
7	0	0	1	0	C	1	1	0	0	L	L	H	L	L	H	1835
8	0	0	1	0	C	1	1	1	0	L	L	H	L	H	L	1835
9	0	0	0	1	C	1 1/	1 1/	0	0	H	L	L	H	H	H	0000
10	0	1	0	0	C	1	1	0	0	H	L	L	H	H	H	0000
11	0	1	0	0	C	1	1	0	0 2/	H	L	L	L	H	L	0830
12	0	1	0	0	C	1	1	0	0 2/	L	H	L	L	H	L	0830
13	0	1	0	0	C	1	1	0	0 2/	H	H	L	L	L	H	0830
14	0	1	0	0	C	1	1	1	0	H	H	L	L	H	L	0830
15	0	0	0	1	C	1 1/	1 1/	0	0	H	L	L	H	H	H	0000
16	0	0	1	0	C	1	1	0	0	H	L	L	H	H	H	0000
17	0	0	1	0	C	1	1	0	0 2/	H	L	L	L	L	H	9995 4/
18	0	0	1	0	C	1	1	1	0	L	H	L	L	H	L	9995

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TABLE III (continued)

ITEM	INPUT PINS OF J4									OUTPUT PINS OF J4						RANGE READOUT 3/ (NIXIE) (±10 meters)
	h	e	f	g	j	k	m	n	s	P	R	S	T	p	q	
19	0	0	1	0	C	1	1	0	0 2/	L	H	L	L	L	H	1835
20	0	0	1	0	C	1	1	1	0	L	H	L	L	H	L	1835
21	0	0	0	1	C	1 1/	1 1/	0	0	H	L	L	H	H	H	0000
22 5/	0	0	0	1	C	1	1	0	0	H	L	L	H	H	H	4000
23 5/	0	0	0	1	C	1	1	0	0 2/	H	L	L	L	L	H	4830
24 5/	0	0	0	1	C	1	1	1	0	H	L	L	L	L	H	4830
25	0	0	0	1	C	1 1/	1 1/	0	0	H	L	L	H	H	H	0000
26 6/	0	0	0	1	C	1	1	0	0	H	L	L	H	H	H	8000
27 6/	0	0	0	1	C	1	1	0	0 2/	H	L	L	L	L	H	8830
28 6/	0	0	0	1	C	1	1	1	0	H	L	L	L	L	H	8830
29	0	0	0	1	C	1 1/	1 1/	0	0	H	L	L	H	H	H	0000

1/ Logical zero momentarily applied.

2/ Logical one momentarily applied.

3/ These range indications shall also appear on the test set when connected to the data line to computer. Disregard test set range indications above 2835.

4/ Logical zero to be momentarily applied to J4-k and J4-m and item 17 repeated if range indication is 0000

5/ Also J4-M shorted to J4-X.

6/ Also J4-N shorted to J4-X.

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TABLE IV

ITEM	INPUT PINS OF J4									OUTPUT PINS	
	h	e	f	g	j	k	m	n	s	J1-P	J1-R
1	0	0	0	1	C	1 1/	1 1/	0	0	28.5 ±2.5 Vdc	0 ±0.5 Vdc
2	0	0	1	0	C	1	1	0	0	28.5 ±2.5 Vdc	0 ±0.5 Vdc
3	0	0	1	0	C	1	1	0	0 2/	28.5 ±2.5 Vdc	0 ±0.5 Vdc
4	0	1	0	0	C	1	1	0	0	28.5 ±2.5 Vdc	0 ±0.5 Vdc
5	0	1	0	0	C	1	1	1	0	0 ±0.5 Vdc	28.5 ±0.5 Vdc

1/ Logical zero momentarily applied.
2/ Logical one momentarily applied.

TABLE V

INPUTS	
PINS OF J4	LOGIC LEVELS
h	0
e	0
f	0
g	1
j	C
k	1 1/
m	1 1/
n	0
s	0

1/ Logical zero momentarily applied.

OUTPUTS	
PINS OF J3	AMPLITUDES
B	4 ±1.5 V (negative polarity)
C	1.4 ±1 V (positive polarity)
F	5.5 ±3 V (negative polarity)
G	+4 ±1.5 Vdc
H	+29 ±3 Vdc
J	+15 ±2 Vdc
K	+5 ±1 Vdc
L	-6 ±1 Vdc
M	+24 ±2 Vdc

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TABLE VI

ITEM	INPUT PINS OF J4									J4-Y OUTPUT
	h	e	f	g	j	k	m	n	s	
1	1	0	0	1	D	1 1/	1 1/	0	0	3/
2	1	0	0	1	D	1	1	0	0 2/	L
3	1	0	0	1	D	1	1	0	0 2/	L
4	1	0	0	1	D	1 1/	1 1/	0	0	3/

1/ Logical zero momentarily applied.
2/ Logical one momentarily applied.
3/ The waveform of Figure 4 shall appear.

TABLE VII

ITEM	MODE	INPUT PINS OF J4										OUTPUT PINS OF J4							RANGE READOUT (NIXIE)
		i	h	e	f	g	j	k	m	n	s	Y	p	q	P	R	S	T	
1	Test	NC	0	0	0	1	C	1 1/	1 1/	0	0	3/	NA	NA	NA	NA	NA	NA	NA
2		NC	0	0	0	1	C	1	1	0	0 2/	3/	NA	NA	NA	NA	NA	NA	NA
3	On	0	1	0	0	1	D	1 1/	1 1/	0	0	NA	H	H	H	L	L	H	0000
4		0	1	0	0	1	D	1	1	0	0 2/	NA	L	H	H	L	L	L	Approx. range
5		0	1	0	0	1	D	1	1	0 2/	0	NA	H	L	H	L	L	L	Approx. range
6		0	1	0	0	1	D	1 1/	1 1/	0	0	NA	H	H	H	L	L	H	0000
7	Auto	NC	1	0	0	1	C	1	1	0	0	NA	H	H	H	L	L	H	0000
8		NC	1	0	0	1	C	1	1	0	0 2/	NA	H	L	H	L	L	L	Approx. range
9		NC	1	0	0	1	C	1 1/	1 1/	0	0	NA	H	H	H	L	L	H	0000
10 4/		NC	1	0	0	1	C	1	1	0	0 2/	NA	L	H	H	L	L	H	9995 5/
11		NC	1	0	0	1	C	1 1/	1 1/	0	0	NA	H	H	H	L	L	H	0000

1/ Logical zero momentarily applied.
2/ Logical one momentarily applied.
3/ Waveform of Figure 4 shall appear.
4/ Receiver aperture shall be covered.
5/ Logical zero to be momentarily applied to J4-k and J4-m and item 10 repeated if range indication is 0000.
NC No Connection
NA Not Applicable

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3.3.1.8 Remote ranging.- With the digital type A signal of Table I applied to each group of input pins for each item as specified in Table VIII, the outputs at J4 and the range (nixie) readout shall be as specified in Table VIII for each item.

TABLE VIII

ITEM	INPUT PINS J4									J1	OUTPUT PINS OF J4						RANGE READOUT (NIXIE)
	i	h	e	f	g	k	m	n	s	U	p	q	P	R	S	T	
1	0	1	0	0	1	1 1/	1 1/	0	0	0	H	H	H	L	L	H	0000
2	0	1	0	0	1	1	1	0	0	0 3/	L	H	H	L	L	L	Approx. range
3	0	1	0	0	1	1	1	0 2/	0	0	H	L	H	L	L	L	Approx. range
4	0	1	0	0	1	1 1/	1 1/	0	0	0	H	H	H	L	L	H	0000

1/ Logical zero momentarily applied.
2/ Logical one momentarily applied.
3/ +8 ±0.5 V momentarily applied.

3.3.1.9 Peak power.- The peak optical output power of the receiver-transmitter shall be not less than one megawatt.

3.3.1.10 Ranging.-

(a) Minimum range. The minimum range shall be 200 meters.
(b) Maximum range. The maximum range shall be 4000 meters.
(c) Range accuracy. The range accuracy shall be plus or minus 10 meters.
(d) Range resolution. The range resolution shall be 20 meters.
(e) Minimum time between ranging. With the type A and type B signals of Table I applied to each group of input pins for each item as specified in Table IX, and each item performed in sequence, the duration of the output voltage at J4-Y for item 2, Table IX shall be no longer than 6 seconds.

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TABLE IX

ITEM	INPUT PINS OF J4									J4-Y OUTPUT
	i	h	e	f	g	k	m	n	s	
1	0	1	0	0	1	1 1/	1 1/	0	0	3/
2	0	1	0	0	1	1	1	0	0 2/	4/

1/ Logical zero momentarily applied.
2/ Logical one momentarily applied.
3/ The waveform of Figure 3 shall appear.
4/ 4 ±1.6 V shall appear for a duration no longer than 6 seconds.

(f) Maximum sustained ranging. The maximum number of rangings shall be 6 per minute for 2 minutes without deterioration of range accuracy.

(g) Events counter. The events counter shall record one event for each activation of the Q-switch or flash tube.

3.3.1.11 Receiver-transmitter leakage. With the receiver-transmitter pressurized internally to at least 9.5 psig, and the relief valve closed, the leakage rate shall not exceed 0.2 psig per hour.

3.3.2 Environmental.

3.3.2.1 Storage temperature.- The receiver-transmitter shall meet the requirements of 3.3.1.1 through 3.3.1.10 at ambient temperature (60°F to 90°F) after exposure and thermal stabilization at -80°F and +160°F.

3.3.2.2 Operating temperatures.- The receiver-transmitter shall meet the requirements of 3.3.1.1 through 3.3.1.10, except 3.3.1.1.2, 3.3.1.1.3, 3.3.1.2, and 3.3.1.5, while exposed to and stabilized at -40°F and +125°F, subsequent to which it shall meet the requirements of 3.3.1.1 through 3.3.1.10 at ambient temperature (60°F to 90°F).

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3.3.2.3 Vibration.- The receiver-transmitter shall meet the requirements of 3.3.1.3 during exposure to the vibratory conditions defined below and shall meet the requirements of 3.3.1.1 through 3.3.1.10 after vibration exposure.

(a)	Amplitude	0.5 inch maximum
(b)	Acceleration	4g maximum
(c)	Crossover frequency	12.5 Hertz (Hz)
(d)	Sweep time	5 to 500 to 5 Hz in 15 minutes, two sweeps in each of three mutually perpendicular axes.

3.3.2.4 Shock.- The receiver-transmitter shall meet the requirements of 3.3.1.1 through 3.3.1.10 after being exposed to a total of six shock impulses applied as follows:

(a) Two shocks of 250g peak amplitude along the longitudinal axis directed toward the receiver-transmitter window
(b) One shock of 175g peak amplitude in each direction of the two transverse axes.

Each shock impulse shall be a half-sine wave or sawtooth with a time duration of 6 milliseconds.

3.3.2.5 Immersion.- The receiver-transmitter shall show no evidence of leakage (gas bubbles from inside the unit) when immersed in water, at 110 ±10°F, to a depth of 2 feet maximum for a 30-minute observation period. Prior to immersion, the electronic section of the receiver-transmitter shall have been pressurized to an internal pressure of 5 ±1 psig.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.1.1 General provisions.- The component and subassembly inspection requirements of MIL-F-13926 form a part of the quality assurance provisions of this specification. Definitions of inspection terms shall be as listed in MIL-STD-109.

4.2 First article (initial production) approval.- The requirement for first article approval and the responsibilities (Government or contractor) for first article testing shall be as specified in the contract. The sample for first article approval tests shall consist of three receiver-transmitters. The sample shall be manufactured in the same manner, using the same materials, equipment, processes, and procedures as used in regular production. All parts and materials, including packaging and packing, shall be obtained from the same source of supply as used in regular production.

4.2.1 Government testing.- When the Government is responsible for conducting first article approval tests, the contractor, prior to submitting the sample to the Government, shall inspect the sample to ensure that it conforms to all the requirements of the contract and submit a record of this inspection with the sample, including certificates of conformance for materials.

4.2.2 Contractor testing.- When the contractor is responsible for conducting first article approval tests, the sample shall be inspected by the contractor for all the requirements of the contract. The sample and a record of this inspection, including certificates of conformance for materials, shall be submitted to the Government for approval. The Government reserves the right to witness the contractor's inspection.

4.3 Inspection provisions.

4.3.1 Submission of product.- Unless otherwise specified by the contracting officer, inspection lot size, lot formation and presentation of lots shall be in accordance with "Submission of Product" provisions of MIL-STD-105.

4.3.2 Examination and tests.

4.3.2.1 Components and subassemblies.- All components and subassemblies shall be inspected in accordance with the inspection provisions contained in Supplementary Quality Assurance Provisions (SQAP) listed in the technical data package (TDP). In the absence of SQAP's, the applicable Quality Assurance Provisions of MIL-F-13926 shall apply.

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4.3.2.2 Final acceptance inspection.- Subsequent to first article approval, examination and tests related to Section 3 herein shall be performed on a single defect (individual characteristic) basis in accordance with MIL-STD-105 and the sampling plans specified in Tables X, XI, and XII herein. Examination and test for packaging and marking shall be in accordance with MIL-P-14232 and Section 5 herein. The tabulated classification of defects in Tables X, XI, and XII shall constitute the minimum inspection to be performed by the supplier after first article approval and prior to Government acceptance or rejection by item or lot.

4.3.2.3 Functional tests.- The requirements and tests of Table X shall be inspected on a 100% basis. All examinations and tests shall be conducted at the standard ambient temperature (60°F to 90°F).

TABLE X

(100 Percent Inspection)

	CHARACTERISTIC	REQUIREMENT	TEST PROCEDURE
101	Laser ON	3.3.1.1.1	4.6.2.1
102	Safety interlock	3.3.1.1.2	4.6.2.2
103	Emergency power switch	3.3.1.1.3	4.6.2.3
104	Low voltage	3.3.1.1.4	4.6.2.4
105	Gunners control	3.3.1.1.5	4.6.2.5
106	Data line (to computer)	3.3.1.2	4.6.3
107	Target selection and reset	3.3.1.3	4.6.4
108	Command signal output (to computer)	3.3.1.4	4.6.5
109	Test access output	3.3.1.5	4.6.6
110	AGC inhibit	3.3.1.5.1	4.6.6.1
111	PFN charge and laser firing	3.3.1.6	4.6.7
112	Laser mode control	3.3.1.7	4.6.8
113	Remote ranging	3.3.1.8	4.6.9
114	Peak power	3.3.1.9	4.6.10
115	Ranging	3.3.1.10	4.6.11
116	Receiver-transmitter leakage	3.3.1.11	4.6.12

4.3.3 Acceptance and rejection.- Rejected lots shall be screened for all defective characteristics. Removal or correction of defective units and resubmittance of rejected lots shall be in accordance with "Acceptance and Rejection" as specified in MIL-STD-105.

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4.4 Special sampling.

4.4.1 General.- One receiver-transmitter shall be selected at random by a Government representative as a control sample from each 100 produced. The control sample shall meet the requirements and test in Table XI.

TABLE XI

CONTROL TEST	REQUIREMENT	TEST PROCEDURE
301 Fabrication	3.1	Applicable drawings - Visual
302 General specification	3.2	MIL-F-13926 - Visual

4.4.2 Environmental.- Three receiver-transmitters shall be selected at random by a Government representative as a control sample from each 50 produced or from each month's production, whichever occurs first. The control samples shall meet the requirements and tests in Table XII.

TABLE XII

CONTROL TEST	REQUIREMENT	TEST PROCEDURE
303 Storage temperature	3.3.2.1	4.6.13.1
304 Operating temperature	3.3.2.2	4.6.13.1
305 Vibration	3.3.2.3	4.6.13.2
306 Shock	3.3.2.4	4.6.13.3
307 Immersion	3.3.2.5	4.6.13.4

4.4.3 Failure of sample.- Should any one item of a special sampling fail to meet the specified test requirements, acceptance of the product shall be suspended by the Government until necessary corrections have been made by the contractor and the resubmitted item has been approved (see 4.3.3).

4.5 Inspection equipment.- Except as otherwise provided for by the contract, the contractor shall supply and maintain inspection equipment in accordance with the applicable requirements of MIL-I-45607.

4.5.1 Government furnished inspection equipment.- Where the contract provides for Government furnished test equipment, supply and maintenance of test equipment shall be in accordance with the applicable requirements specified in MIL-I-45607.

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4.5.2 Contractor furnished inspection equipment.

4.5.2.1 Government design.- Unless otherwise specified in the contract, all inspection equipment specified by drawing number in specifications or SQAP forming a part of the contract shall be supplied by the contractor in accordance with applicable technical data listed in the TDP.

4.5.2.2 Contractor design.- The contractor shall design and supply inspection equipment compatible with the "Test Methods and Procedures" specified in 4.6 of this specification and with the component inspection procedures specified in "Examination" and "Test Facilities" requirements of MIL-F-13926. Since tolerance of test equipment is normally considered to be within 10% of the product tolerance for which it is intended, this inherent error in the test equipment design must be considered as part of the prescribed product tolerance limit. Thus, concept, construction, materials, dimensions, and tolerances used in the design of test equipment shall be so selected and controlled as to ensure that the test equipment will reliably indicate acceptability of a product which does not exceed 90% of the prescribed tolerance limit and permit positive rejection when nonconforming. Construction shall be such as to facilitate routine calibration of test equipment.

4.5.3 Test equipment.- In conjunction with 4.5.2, the standard test equipment listed in Table XIII shall be utilized in performing the applicable test of 4.6.

TABLE XIII

	NOMENCLATURE	DESCRIPTION
1.	D-C Power Supply	Plus 24 ±2 volts Ripple: 0.1 volt rms maximum Source impedance: <0.01 ohm
2.	D-C Voltmeter	Range: 0 to 50 volts Accuracy: ±0.3%
3.	Ammeter	Range: 0 to 100 microamperes Accuracy ±1%
4.	Oscilloscope with Camera	Frequency response: dc to 150 MHz Sensitivity: 5 mV/cm Sweep time: 0.01 μsec/cm Camera lens: f/1.3 (Tektronix C40 or equivalent)

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TABLE XIII (continued)

NOMENCLATURE	DESCRIPTION
5. Thermo Pile	TRG 108, Tekmar Instruments, Inc. or equivalent
6. Energy Meter	TRG 113, Tekmar Instruments, Inc. or equivalent
7. Signal Source	Capable of providing d-c level as specified in Table I, item 1.3
8. Test Set	Provides decimal readout on nixie tubes from 8-4-2-1 binary coded decimal inputs.
9. Target No. 1	12-inch minimum diameter circle, 15 percent minimum diffuse reflectivity and located, within ±1 meter, at 215 ±5 meters from window of receiver-transmitter.
10. Target No. 2	16-inch minimum diameter circle, 15 percent minimum diffuse reflectivity and located, within ±1 meter, at the actual distance of Target No. 3 minus 20 meters from window of receiver-transmitter.
11. Target No. 3	4 feet square minimum, 15 percent minimum diffuse reflectivity unless otherwise specified and located, within ±1 meter, at 500 ±15 meters from window of receiver-transmitter.

4.6 Test methods and procedures.

4.6.1 Test conditions. - Unless otherwise specified, the test conditions shall be in accordance with the "Test Facilities" requirements of MIL-F-13926.

4.6.1.1 Initial conditions. - Connect the loads, auxiliary equipment, power, and input signals to the receiver-transmitter as specified in Table I. Except for the power control test, the emergency power switch shall be in the OFF position.

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4.6.2 Power controls.

4.6.2.1 Laser On test.- Apply the logical zero type A signal of Table I to J4-s and J4-h, then apply the Laser On signal of Table I as specified. Measure the voltages at the output pins listed in Table II and observe the waveform at J4-Y. The output voltages and waveform shall meet the requirements of 3.3.1.1.1.

4.6.2.2 Safety interlock test.- Remove the receiver-transmitter cover and measure the voltage at J2-H. The requirements of 3.3.1.1.2 shall be met.

4.6.2.3 Emergency power switch test.- Remove the Laser On and Gunners Control signals, and sequence the Emergency Power switch (S2) through its positions. The requirements of 3.3.1.1.3 shall be met.

4.6.2.4 Low voltage test.- Remove the 24 Vdc from J2-J and observe the voltage at J3-J. The requirement of 3.3.1.1.4 shall be met.

4.6.2.5 Gunners control test.- Apply the Gunners Control signal of Table I as specified and measure the voltage at J4-r. The requirement of 3.3.1.1.5 shall be met.

4.6.3 Data line (to computer) test.- Remove the Counter Circuit Card (10559305) and replace it with a card connected as shown on Figure 3, or equivalent. Open all switches on the test card and measure the output voltages at J1-A through J1-L. Close the switches singly or in sequence and measure the output voltages at J1-A through J1-L. The requirements of 3.3.1.2 shall be met.

4.6.4 Target selection and reset test.- Apply the type A and type B signals of Table I to the group of input pins for each item in sequence as specified in Table III. Measure the digital signal at each output pin specified for that item and observe the range readout specified for that item in Table III. The requirements of 3.3.1.3 shall be met.

4.6.5 Command signal output (to computer).- Apply the type A and type B signals of Table I to the group of input pins for each item in sequence as specified in Table IV. Measure the digital signal at each output pin specified for items, 1, 3, and 5 in Table IV. The requirements of 3.3.1.4 shall be met.

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4.6.6 Test access output.- Enable the test oscillator (by removing large cover) and apply the type A and type B signals of Table I simultaneously to the input pins as specified in Table V, followed by a logical zero type A signal momentarily applied to J4-k and J4-m. Measure the voltage amplitudes at the output pins specified in Table V. The requirements of 3.3.1.5 shall be met.

4.6.6.1 AGC inhibit test.- Under the conditions of 4.6.6, cover the receiver-transmitter window and measure the voltage at J3-D. The requirement of 3.3.1.5.1 shall be met.

NOTE: PFN voltage adjustment. Prior to operation, initially adjust R10 of the PFN charge power supply (10559345) for the PFN voltage determined by test and recorded for the transmitter component assembly (10559479), in accordance with MIL-C-14880. Readjust the PFN voltage for a laser energy output of 50 ±5 millijoules.

Ranging operation. The input logic signals to the receiver-transmitter for laser firing shall be as specified in Table IX.

4.6.7 PFN charge and laser firing test.- Apply the type A and type B signals of Table I to the group of input pins for each item, as specified in Table VI and perform each item in sequence. Observe the output at J4-Y for each item of Table VI. The requirements of 3.3.1.6 shall be met.

4.6.8 Laser mode control test.- Apply the type A and type B signals of Table I to the group of input pins for each item as specified in Table VII and perform each item in sequence. Measure the voltage or observe the waveform, as applicable, at each output pin specified for that item and observe the range readout specified for that item in Table VII. The requirements of 3.3.1.7 shall be met.

4.6.9 Remote ranging test.- Apply the type A signal of Table I to the group of input pins for each item as specified in Table VIII. Measure the digital signal at each output pin specified for that item and observe the range readout specified for that item in Table VIII. The requirements of 3.3.1.8 shall be met.

4.6.10 Peak power test.- Measure the peak optical output power of the receiver-transmitter during a firing operation. The requirement of 3.3.1.9 shall be met.

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4.6.11 Ranging test.

4.6.11.1 Minimum range.- Perform five ranging operations against target No. 1 specified in Table XIII and observe the range (nixie) readouts. The readouts shall be the actual target range within the specified range accuracy, thereby meeting the requirement of 3.3.1.10(a).

4.6.11.2 Maximum range.- Connect J3-C to ground and place neutral density filters over the receiver aperture with an attenuation of 42 ±1 db. Perform ten ranging operations against target No. 3 specified in Table XIII. The target shall be flat, normal to the laser beam within ±10 degrees, with a minimum diffuse reflectivity of 80%. Observe the range (nixie) readouts. The readouts shall be the actual target range within the specified range accuracy, thereby meeting the requirement of 3.3.1.10(b).

4.6.11.3 Range accuracy.- Perform five ranging operations against target No. 3 specified in Table XIII and observe the range (nixie) readouts. The accuracy of the indicated range shall meet the requirement of 3.3.1.10(c).

4.6.11.4 Range resolution.- Perform five ranging operations against targets No. 2 and No. 3 specified in Table XIII. The targets shall be aligned so as to allow the transmitted laser beam to strike both targets. Observe the range (nixie) readouts. The readouts shall be the actual ranges to the targets within the specified range accuracy, thereby meeting the requirement of 3.1.1.10(d).

4.6.11.5 Minimum time between ranging.- Apply the type A signal of Table I to the group of input pins for each item as specified in Table IX, in sequence and observe the output at J4-Y. The requirement of 3.3.1.10(e) shall be met.

4.6.11.6 Maximum sustained ranging.- Perform twelve ranging operations within a 2-minute period against target No. 1 specified in Table XIII. (This may be performed concurrently with the test of 4.6.11.1.) Observe the indicated range accuracy. The requirement of 3.3.1.10(f) shall be met.

4.6.11.7 Events counter.- Observe the performance of the events counter during one ranging operation. The requirement of 3.3.1.10(g) shall be met.

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4.6.12 Receiver-transmitter leakage test. - With the unit unpressurized, connect the fill valve to a nitrogen cylinder through suitable control valves and pressure gauges and pressurize to 14.5 ±1 psig. Slowly reduce the input pressure until the relief valve closes. The leakage rate, observed for a minimum period of 15 minutes, shall meet the requirements of 3.3.1.11. (Corrections may be made in the observed pressure reading for significant temperature changes during the test.)

4.6.12.1 Receiver-transmitter purging and pressurization. - With the unit attached to a nitrogen cylinder, complete the following purge operations:

(a) Pressurize the laser head section at an inlet pressure of 14.5 ±1 psig and allow it to purge for a minimum period of 1 minute, then depressurize to ambient pressure. Repeat this procedure at least four times.
(b) Repressurize the laser head section with an inlet pressure of 14.5 ±1 psig and allow it to purge for a minimum of 20 minutes; then reduce the internal pressure to less than 3 psig. Repressurize again to a pressure of 10.5 ±1 psig.

4.6.13 Environmental tests. - The test equipment shall be in accordance with the "Test Facilities" requirements of MIL-F-13926 except that the receiver-transmitter may be mounted on an inside surface of the test chamber for the temperature test and the tolerance requirements for vibration frequency measuring instruments shall not apply.

4.6.13.1 Temperature test. -

NOTE: The rate of temperature change shall be less than 7°F per minute during temperature cycling.

(a) Place the receiver-transmitter in the test chamber and reduce the chamber temperature to -80°F.
(b) Soak the receiver-transmitter at this temperature for at least 5 hours with the unit not operating.
(c) Raise the chamber temperature to -40°F.
(d) Following thermal stabilization at -40°F, place the receiver-transmitter in operation and perform the tests specified in 4.6.2.1 through 4.6.11, except

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4.6.2.2, 4.6.2.3, 4.6.3, and 4.6.6. During the range accuracy test, 4.6.10.2, the target location shall be between 450 and 2000 meters from the receiver-transmitter.

(e) Raise the chamber temperature to +160°F.
(f) Soak the receiver-transmitter at +160°F for at least 5 hours with the assembly not operating.
(g) Reduce the chamber temperature to +125°F.
(h) Following thermal stabilization at +125°F, place the receiver-transmitter in operation and perform the tests specified in 4.6.2.1 through 4.6.11, except 4.6.2.2, 4.6.2.3, 4.6.3, and 4.6.6. During the range accuracy test, 4.6.10.2, the target location shall be between 450 and 2000 meters from the receiver-transmitter.
(i) Reduce the chamber temperature to ambient.
(j) Following stabilization at ambient temperature, place the receiver-transmitter in operation and perform the test specified in 4.6.2.1 through 4.6.11.

The receiver-transmitter shall meet the requirements of 3.3.2.1 and 3.3.2.2.

4.6.13.2 Vibration.- With the receiver-transmitter firmly mounted to the vibration table, perform the following:

(a) Test in accordance with Method 514-I of MIL-STD-810 for category (f) equipment, except that the acceleration shall not exceed ±4g and the maximum excursion (double amplitude) shall be 0.5 inch.
(b) Sweep the frequency from 5 to 500 Hz and back to 5 Hz in a period of 15 minutes. Perform two sweeps in each of the three mutually perpendicular axes.
(c) During the vibration exposure, perform the tests specified in 4.6.4.
(d) Following vibration cycling, perform a visual-tactile examination for damage and loose parts.
(e) Place the receiver-transmitter in operation and perform the tests specified in 4.6.2.1 through 4.6.11.

The receiver-transmitter shall meet the requirements of 3.3.2.3.

4.6.13.3 Shock.- With the receiver-transmitter rigidly mounted to the shock machine table, perform the following:

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(a) Test in accordance with Method 516, procedure IV of MIL-STD-810 with the following exceptions:

(1) Apply two shock impulses, each with a 250g peak amplitude, half-sine wave or sawtooth, and 6 milliseconds duration, along the longitudinal axis directed toward the receiver-transmitter window.
(2) Apply one shock impulse with a 175g peak amplitude, half-sine wave or sawtooth, and 6 milliseconds duration, along each direction of the two transverse axes.

(b) After exposure to shock, perform a visual-tactile examination for damage and loose parts.
(c) Place the receiver-transmitter in operation and perform the tests specified in 4.6.2.1 through 4.6.11.

The receiver-transmitter shall meet the requirements of 3.3.2.4.

4.6.13.4 Immersion.- Connect the leak test port of the receiver-transmitter to a cylinder of nitrogen or air and pressurize it to 5 ±1 psig. Submerge the unit in water at 110° ±10°F to a depth not to exceed 2 feet and observe the unit for evidence of leakage (gas bubbles) for a period of 30 minutes. The unit shall meet the requirements of 3.3.2.5.

NOTE: The unit shall not be left unattended during immersion.

5. PREPARATION FOR DELIVERY

5.1 Packaging, packing, and marking.- Packaging, packing, and marking shall be in accordance with Packaging Data Sheet 10559281. The level of protection shall be as specified in the procurement document.

6. NOTES

6.1 Intended use.- The receiver-transmitter is a unit of the Laser Range Finder AN/VVS-1. This unit contains the laser transmitter, power supplies, energy storage network, laser receiver including basic ranging circuitry, and alignment optics.

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6.2 Ordering data.- Procurement documents should specify the following:

(a) Title, number, and date of this specification
(b) Applicable packaging data sheet number (see 5.1)
(c) Selection of applicable levels of preservation, packaging, and packing
(d) Applicable stock number
(e) Requirement for First Article Submission.

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Preparing Activity:

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Project No. 1240-A134

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

[ ] YES [ ] NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

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